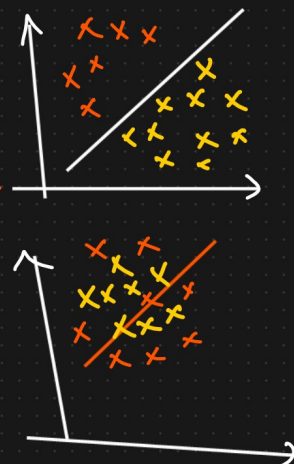


Agenda

- ① Naive Bayes Algorithm (Classification Algorithm) ✓
- ② Prerequisites { Linear Regression, Ridge, Lasso, Elastic Net, Logistic } ✓
- ③ KNOWN ✓



Assignment

9
KNOWN

① Naive Bayes Intuition

Rolling a Dice { 1, 2, 3, 4, 5, 6 }

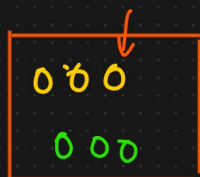
$$P(1) = \frac{1}{6}$$

$$P(2) = \frac{1}{6}$$

$$P(3) = \frac{1}{6}$$

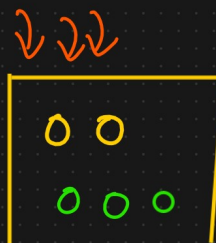
Independent Events

Dependent Events



First Event

$$P(y) = \frac{3}{6} = \frac{1}{2} \quad \checkmark$$



$$\rightarrow P(u/y) = \frac{3}{5}$$

Dependent

$$P(y \text{ and } u) = P(y) * P(u/y)$$

$$= \frac{1}{2} * \frac{3}{5} = \frac{3}{10}$$

$$P(A \text{ and } B) = P(A) * P(B/A)$$

↑ ↑

A and B

u and y

Naive Bayes
↑↑
Conditional Probability

Dependent events
↑

$$P(A \text{ and } B) = P(B \text{ and } A)$$

↓

↓

$$P(A) * P(B/A) = P(B) * P(A/B)$$

$$P(B/A) = \frac{P(B) * P(A/B)}{P(A)}$$

Bay's Theorem

$$\begin{matrix} 000 \\ 00 \end{matrix} \quad P(w)$$

$$P(0|w)$$

$$\begin{matrix} 000 \\ 00 \end{matrix} \quad P(0)$$

$$P(w|0)$$

Dataset

I/P features

$$x_1 \ x_2 \ x_3 \ x_4 \ \dots \ x_n \quad y \text{ O/P}$$

$$P(y/x_1, x_2, x_3, \dots, x_n) = \frac{P(y) * P(x_1, x_2, x_3, \dots, x_n/y)}{P(x_1, x_2, x_3, \dots, x_n)}$$

$$= \frac{P(y) * P(x_1/y) * P(x_2/y) * P(x_3/y) \dots * P(x_n/y)}{P(x_1) * P(x_2) * P(x_3) * \dots * P(x_n)}$$

Binary dataset

$x_1 \quad x_2 \quad x_3 \quad \text{O/P}$
Yes
No

$$P(x_1, x_2, x_3 / \text{Yes})$$

$$P(y = \text{Yes} / x_i) = \frac{P(y) * P(x_1/y) * P(x_2/y) * P(x_3/y)}{P(x_1) * P(x_2) * P(x_3)}$$

Constant Ignore $\rightarrow$ $P(x_1) * P(x_2) * P(x_3)$ $\leftarrow$ # fixed

$$P(y = No | x_i) = P(No) * P(x_1/No) * P(x_2/No) * P(x_3/No)$$

Constant $\rightarrow$ ~~$P(x_1) * P(x_2) * P(x_3)$~~ $\leftarrow$ # fixed
Ignore

Let say

$$P(Yes/x_i) = 0.13$$

$$P(No/x_i) = 0.05$$

$$P(Yes/x_i) = \frac{0.13}{0.13 + 0.05} = \frac{0.13}{0.18} = 72\% \quad \begin{matrix} \geq 50\% \\ < 50\% \end{matrix}$$

$$P(No/x_i) = \frac{0.05}{0.13 + 0.05} = 0.28 = 28\%$$

$\Rightarrow 100\%$

DATASET

Naive Bayes

Outlook

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

	Yes	No	P(y)	P(n)
Sunny	2	3	$\frac{2}{9}$	$\frac{3}{5}$
Overcast	4	0	$\frac{4}{9}$	$\frac{0}{5}$
Rain	3	2	$\frac{3}{9}$	$\frac{2}{5}$
	9	5		

Temperature (Overcast, Cool)

	Yes	No	P(y)	P(n)
Hot	2	2	$\frac{2}{9}$	$\frac{2}{5}$
Mild	4	2	$\frac{4}{9}$	$\frac{2}{5}$
Cold	3	1	$\frac{3}{9}$	$\frac{1}{5}$
	9	5		

PLAY

	Yes	No	P(Yes)	P(No)
Yes	9		$\frac{9}{14}$	$\frac{5}{14}$
No		5		
	14			

$\rightarrow$ (Sunny, Hot) $\rightarrow$ O/p Naive Bayes $\Rightarrow$ % No

$$P(\text{Yes} | (\text{Sunny}, \text{Hot})) = \frac{P(\text{Yes}) * P(\text{Sunny} | \text{Yes}) * P(\text{Hot} | \text{Yes})}{\cancel{P(\text{Sunny})} * \cancel{P(\text{Hot})}} = \frac{\cancel{9}}{1479} * \frac{2}{9} * \frac{2}{9}$$

$$P(\text{No} | (\text{Sunny}, \text{Hot})) = \frac{P(\text{No}) * P(\text{Sunny} | \text{No}) * P(\text{Hot} | \text{No})}{\cancel{P(\text{Sunny})} * \cancel{P(\text{Hot})}} = \frac{2}{63} = 0.031$$

$$= \frac{\cancel{9}}{147} * \frac{3}{5} * \frac{21}{8}$$

$$= \frac{3}{35} = 0.08571$$

Real Probability

$$P(\text{Yes} | (\text{Sunny}, \text{hot})) = 0.031$$

$$P(\text{Yes} | (\text{Sunny}, \text{hot})) = \frac{0.031}{0.031 + 0.0857}$$

$$= 27\%$$

$$P(\text{No} | (\text{Sunny}, \text{Hot})) = 0.0857$$

$$= 27\%$$

$$P(\text{No} | (\text{Sunny}, \text{hot})) = \frac{1 - 0.27}{1} = 73\%$$

→ DATASET
1000

X

Y

x_1, x_2, x_3, x_4

0/p

NL Model

↓

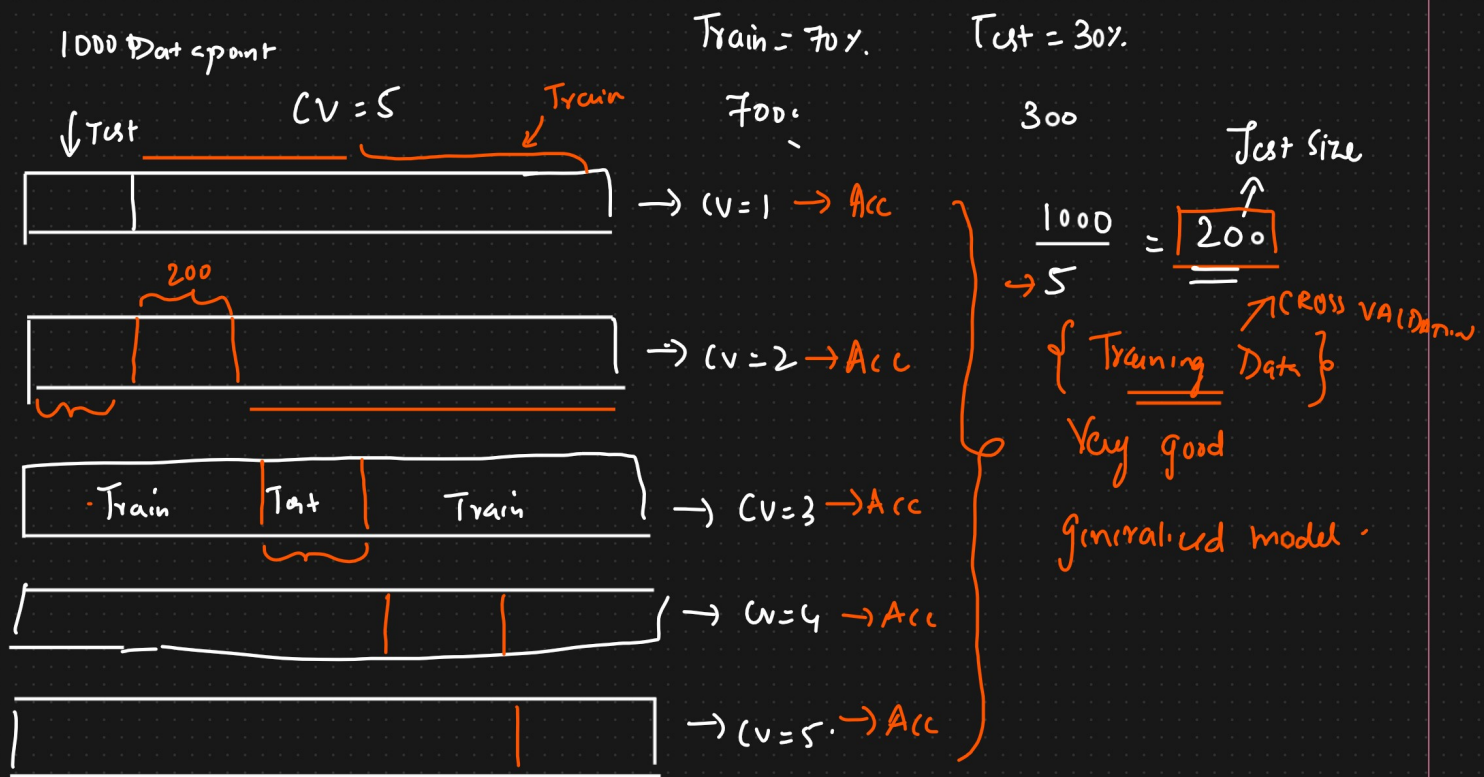
Jest Data

↓

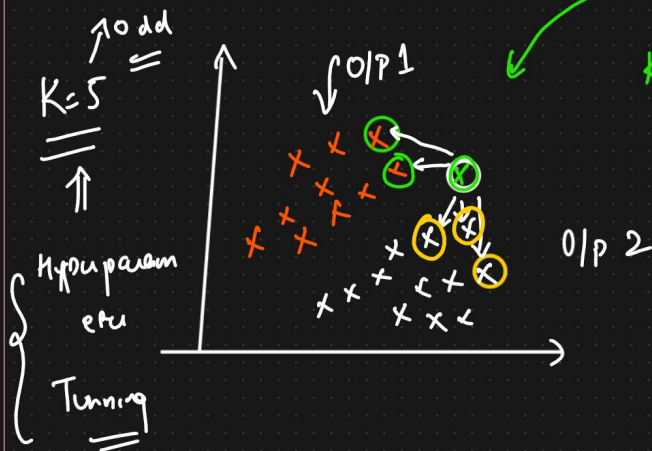
{ Train Test Split }

Dataset { Train ← 70%,
Jest ← 30%

CROSS VALIDATION



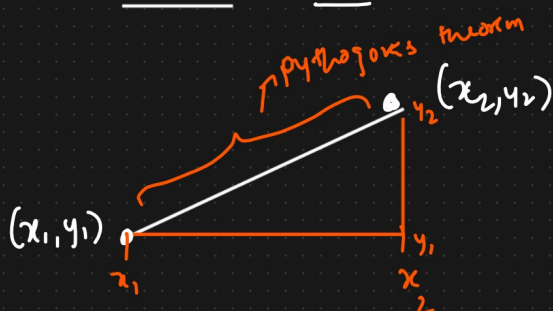
⑧ K Nearest Neighbour (KNN) → Classification and Regression Problem



① Euclidean Distance

② Manhattan Distance.

① Euclidean Distance



$$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$